

Chapter 6: KKT Conditions

MATH60005/70005: Optimisation
Term 1, Academic Year 25-26

Dr Dante Kalise
Department of Mathematics
Imperial College London
dkaliseb@ic.ac.uk

Separation Theorem	1
KKT Conditions for Linearly Constrained Problems	3
Orthogonal projections	5
KKT conditions for nonlinear problems	6
KKT conditions for nonlinear convex problems	7

Separation Theorem

To begin our study of optimality conditions for linearly constrained problems we need first some technical results, known as *Alternative and Separation Theorems*.

A hyperplane

$$H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^\top \mathbf{x} = b\} \quad (\mathbf{a} \in \mathbb{R}^n \setminus \{\mathbf{0}\}, b \in \mathbb{R})$$

is said to strictly separate a point $\mathbf{y} \notin S$ from S if

$$\mathbf{a}^\top \mathbf{y} > b,$$

and

$$\mathbf{a}^\top \mathbf{x} \leq b \text{ for all } \mathbf{x} \in S.$$

Theorem (Separation of a Point from a Closed and Convex Set). *Let $C \subseteq \mathbb{R}^n$ be a nonempty closed and convex set, and let $\mathbf{y} \notin C$. Then there exists $\mathbf{p} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ and $\alpha \in \mathbb{R}$ such that $\mathbf{p}^\top \mathbf{y} > \alpha$ and $\mathbf{p}^\top \mathbf{x} \leq \alpha$ for all $\mathbf{x} \in C$.*



Proof. By the second orthogonal projection theorem, the vector $\bar{\mathbf{x}} = P_C(\mathbf{y}) \in C$ satisfies

$$(\mathbf{y} - \bar{\mathbf{x}})^\top (\mathbf{x} - \bar{\mathbf{x}}) \leq 0 \quad \text{for all } \mathbf{x} \in C,$$

which is the same as

$$(\mathbf{y} - \bar{\mathbf{x}})^\top \mathbf{x} \leq (\mathbf{y} - \bar{\mathbf{x}})^\top \bar{\mathbf{x}} \quad \text{for all } \mathbf{x} \in C.$$

Denote $\mathbf{p} = \mathbf{y} - \bar{\mathbf{x}} \neq \mathbf{0}$ and $\alpha = (\mathbf{y} - \bar{\mathbf{x}})^\top \bar{\mathbf{x}}$. Then,

$$\mathbf{p}^\top \mathbf{x} \leq \alpha \quad \text{for all } \mathbf{x} \in C.$$

On the other hand, we have

$$\mathbf{p}^\top \mathbf{y} = (\mathbf{y} - \bar{\mathbf{x}})^\top \mathbf{y} = (\mathbf{y} - \bar{\mathbf{x}})^\top (\mathbf{y} - \bar{\mathbf{x}}) + (\mathbf{y} - \bar{\mathbf{x}})^\top \bar{\mathbf{x}} = \|\mathbf{y} - \bar{\mathbf{x}}\|^2 + \alpha > \alpha.$$

□

Lemma (Farkas' Lemma - an Alternative Theorem). *Let $\mathbf{c} \in \mathbb{R}^n$ and $\mathbf{A} \in \mathbb{R}^{m \times n}$. Then exactly one of the following systems has a solution:*

I. $\mathbf{Ax} \leq \mathbf{0}, \mathbf{c}^\top \mathbf{x} > 0.$

II. $\mathbf{A}^\top \mathbf{y} = \mathbf{c}, \mathbf{y} \geq \mathbf{0}.$

An alternative formulation is the following:

Lemma (Farkas' Lemma - Second Formulation). *Let $\mathbf{c} \in \mathbb{R}^n$ and $\mathbf{A} \in \mathbb{R}^{m \times n}$. Then the following two claims are equivalent:*

(A) *The implication $\mathbf{Ax} \leq \mathbf{0} \Rightarrow \mathbf{c}^\top \mathbf{x} \leq 0$ holds true.*

(B) *There exists $\mathbf{y} \in \mathbb{R}_+^m$ such that $\mathbf{A}^\top \mathbf{y} = \mathbf{c}$.*

Proof. Suppose that system (B) is feasible. Then, there exists $\mathbf{y} \in \mathbb{R}_+^m$ such that $\mathbf{A}^\top \mathbf{y} = \mathbf{c}$. To see that the implication (A) holds, suppose that $\mathbf{Ax} \leq \mathbf{0}$ for some $\mathbf{x} \in \mathbb{R}^n$. Multiplying this inequality from the left by $\mathbf{y}^\top$ we obtain:

$$\mathbf{y}^\top \mathbf{Ax} \leq 0,$$

and hence,

$$\mathbf{c}^\top \mathbf{x} \leq 0.$$

Now, suppose that the implication (A) is satisfied, and let us show that the system (B) is feasible. Suppose in contradiction that system (B) is infeasible. Consider the following closed and convex set

$$S = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} = \mathbf{A}^\top \mathbf{y} \text{ for some } \mathbf{y} \in \mathbb{R}_+^m\}.$$



Note that $\mathbf{c} \notin S$. By the separation theorem, there exists $\mathbf{p} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ and $\alpha \in \mathbb{R}$ such that $\mathbf{p}^\top \mathbf{c} > \alpha$ and

$$\mathbf{p}^\top \mathbf{x} \leq \alpha \text{ for all } \mathbf{x} \in S.$$

$\mathbf{0} \in S$ implies that $\alpha \geq 0 \Rightarrow \mathbf{p}^\top \mathbf{c} > 0$, and the inequality above is equivalent to

$$\mathbf{p}^\top \mathbf{A}^\top \mathbf{y} \leq \alpha \text{ for all } \mathbf{y} \geq \mathbf{0}$$

or to

$$(\mathbf{A}\mathbf{p})^\top \mathbf{y} \leq \alpha \text{ for all } \mathbf{y} \geq \mathbf{0}.$$

Therefore, $\mathbf{A}\mathbf{p} \leq \mathbf{0}$, which is a contradiction to the assertion that implication (A) holds. $\square$

Example. What does this mean for $\mathbf{A} = \begin{pmatrix} 1 & 5 \\ -1 & 2 \end{pmatrix}$, $\mathbf{c} = \begin{pmatrix} -1 \\ 9 \end{pmatrix}$?

Theorem (Gordan's Alternative Theorem). *Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. Then exactly one of the following two systems has a solution:*

- I. $\mathbf{A}\mathbf{x} < \mathbf{0}$.
- II. $\mathbf{p} \neq \mathbf{0}, \mathbf{A}^\top \mathbf{p} = \mathbf{0}, \mathbf{p} \geq \mathbf{0}$.

KKT Conditions for Linearly Constrained Problems

The Karush-Kuhn-Tucker (or KKT) conditions are a set of fundamental characterizations of the solution of convex optimization problems. Here we study some variants for the linearly constrained case.

Theorem (KKT conditions for Linearly Constrained Problems - Necessary Optimality Conditions). *Consider the minimization problem*

$$\begin{cases} \min f(\mathbf{x}) \\ \text{subject to } \mathbf{a}_i^\top \mathbf{x} \leq b_i, \quad i = 1, 2, \dots, m \end{cases} \quad (\text{LCP})$$

where f is continuously differentiable over $\mathbb{R}^n$, $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m \in \mathbb{R}^n, b_1, b_2, \dots, b_m \in \mathbb{R}$ and let $\mathbf{x}^*$ be a local minimum point of (LCP). Then, there exist $\lambda_1, \lambda_2, \dots, \lambda_m \geq 0$ such that

$$\nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i \mathbf{a}_i = \mathbf{0},$$

and

$$\lambda_i (\mathbf{a}_i^\top \mathbf{x}^* - b_i) = 0, \quad i = 1, 2, \dots, m.$$



Proof. If $\mathbf{x}^*$ is a local minimum, this implies $\mathbf{x}^*$ is a stationary point, meaning

$$\nabla f(\mathbf{x}^*)^\top (\mathbf{x} - \mathbf{x}^*) \geq 0$$

for every $\mathbf{x} \in \mathbb{R}^n$ satisfying $\mathbf{a}_i^\top \mathbf{x} \leq b_i$ for any $i = 1, 2, \dots, m$. Now, denote the set of active constraints by

$$I(\mathbf{x}^*) = \{i : \mathbf{a}_i^\top \mathbf{x}^* = b_i\}.$$

Making the change of variables $\mathbf{y} = \mathbf{x} - \mathbf{x}^*$, we have $\nabla f(\mathbf{x}^*)^\top \mathbf{y} \geq 0$ for any $\mathbf{y} \in \mathbb{R}^n$ satisfying

$$\mathbf{a}_i^\top (\mathbf{y} + \mathbf{x}^*) \leq b_i, \quad i = 1, 2, \dots, m.$$

Or equivalently $\nabla f(\mathbf{x}^*)^\top \mathbf{y} \geq 0$ for any $\mathbf{y}$ satisfying

$$\begin{aligned} \mathbf{a}_i^\top \mathbf{y} &\leq 0 & i \in I(\mathbf{x}^*) \\ \mathbf{a}_i^\top \mathbf{y} &\leq b_i - \mathbf{a}_i^\top \mathbf{x}^* & i \notin I(\mathbf{x}^*). \end{aligned}$$

The second set of inequalities can be removed, that is, we will prove that

$$\mathbf{a}_i^\top \mathbf{y} \leq 0 \text{ for all } i \in I(\mathbf{x}^*) \Rightarrow \nabla f(\mathbf{x}^*)^\top \mathbf{y} \geq 0.$$

For this, suppose that $\mathbf{y}$ satisfies $\mathbf{a}_i^\top \mathbf{y} \leq 0$ for all $i \in I(\mathbf{x}^*)$. Since $b_i - \mathbf{a}_i^\top \mathbf{x}^* > 0$ for all $i \notin I(\mathbf{x}^*)$, it follows that there exists a small enough $\alpha > 0$ for which $\mathbf{a}_i^\top (\alpha \mathbf{y}) \leq b_i - \mathbf{a}_i^\top \mathbf{x}^*$. Thus, since in addition $\mathbf{a}_i^\top (\alpha \mathbf{y}) \leq 0$ for any $i \in I(\mathbf{x}^*)$, it follows by the stationarity condition that $\nabla f(\mathbf{x}^*)^\top \mathbf{y} \geq 0$. Therefore, we have shown $\mathbf{a}_i^\top \mathbf{y} \leq 0$ for all $i \in I(\mathbf{x}^*)$ implies that $\nabla f(\mathbf{x}^*)^\top \mathbf{y} \geq 0$. Now, by Farkas' lemma, there exists $\lambda_i \geq 0, i \in I(\mathbf{x}^*)$ such that

$$-\nabla f(\mathbf{x}^*) = \sum_{i \in I(\mathbf{x}^*)} \lambda_i \mathbf{a}_i.$$

Finally, defining $\lambda_i = 0$ for all $i \notin I(\mathbf{x}^*)$ we get that $\lambda_i (\mathbf{a}_i^\top \mathbf{x}^* - b_i) = 0$ for all $i \in \{1, 2, \dots, m\}$ and

$$\nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i \mathbf{a}_i = \mathbf{0}.$$

□

The previous theorem can be improved to necessary and sufficient conditions when f is convex.

Theorem (KKT Conditions for Convex Linearly Constrained Problems - Necessary and Sufficient Optimality Conditions). *Consider the minimization problem (LCP) where in addition f is a convex continuously differentiable function over $\mathbb{R}^n$, and let $\mathbf{x}^*$ be a feasible solution. Then $\mathbf{x}^*$ is an optimal solution if and only if there exist $\lambda_1, \lambda_2, \dots, \lambda_m \geq 0$ such that*

$$\nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i \mathbf{a}_i = \mathbf{0}$$

and

$$\lambda_i (\mathbf{a}_i^\top \mathbf{x}^* - b_i) = 0, \quad i = 1, 2, \dots, m$$



Proof. Necessity was already proven. For sufficiency, suppose that $\mathbf{x}^*$ is a feasible solution of (LCP) satisfying the optimality conditions. Let $\mathbf{x}$ be a feasible solution of (LCP). Define the function

$$h(\mathbf{x}) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i (\mathbf{a}_i^\top \mathbf{x} - b_i) .$$

The condition $\nabla h(\mathbf{x}^*) = \mathbf{0}$ implies that $\mathbf{x}^*$ is a minimizer of h over $\mathbb{R}^n$. From here, it follows that

$$f(\mathbf{x}^*) = f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i (\mathbf{a}_i^\top \mathbf{x}^* - b_i) \leq f(\mathbf{x}) + \sum_{i=1}^m \lambda_i (\mathbf{a}_i^\top \mathbf{x} - b_i) \leq f(\mathbf{x}) ,$$

that is, $\mathbf{x}^*$ is a minimizer of f . □

We conclude by stating the KKT conditions associated to linear problems with equality and inequality constraints.

Theorem (KKT conditions for Linearly Constrained Problems). *Consider the minimization problem*

$$\begin{cases} \min f(\mathbf{x}) \\ \text{subject to} & \mathbf{a}_i^\top \mathbf{x} \leq b_i, \quad i = 1, 2, \dots, m \\ & \mathbf{c}_j^\top \mathbf{x} = d_j, \quad j = 1, 2, \dots, p \end{cases} \quad (\text{LCPI})$$

where f is continuously differential, $\mathbf{a}_i, \mathbf{c}_j \in \mathbb{R}^n, b_i, d_j \in \mathbb{R}$.

(i) (necessity of the KKT conditions) If $\mathbf{x}^*$ is a local minimum of (LCPI) then there exist $\lambda_1, \lambda_2, \dots, \lambda_m \geq 0$ and $\mu_1, \mu_2, \dots, \mu_p \in \mathbb{R}$ such that

$$\begin{aligned} \nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i \mathbf{a}_i + \sum_{j=1}^p \mu_j \mathbf{c}_j &= \mathbf{0} \\ \lambda_i (\mathbf{a}_i^\top \mathbf{x}^* - b_i) &= 0, \quad i = 1, 2, \dots, m . \end{aligned}$$

(ii) (sufficiency in the convex case) If f is convex over $\mathbb{R}^n$ and $\mathbf{x}^*$ is a feasible solution of (LCPI) for which there exist $\lambda_1, \dots, \lambda_m \geq 0$ and $\mu_1, \dots, \mu_p \in \mathbb{R}$ such that the conditions are satisfied, then $\mathbf{x}^*$ is an optimal solution of (LCPI).

Examples: Solve the problem

$$\begin{aligned} \min & \quad \frac{1}{2} (x_1^2 + x_2^2 + x_3^2) \\ \text{s.t.} & \quad x_1 + x_2 + x_3 = 3 . \end{aligned}$$

Orthogonal projections

Using KKT conditions, show the following results!



Orthogonal Projection onto Affine Spaces

Let C be the affine space $C = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{A}\mathbf{x} = \mathbf{b}\}$, where $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$. Then,

$$P_C(\mathbf{y}) = \mathbf{y} - \mathbf{A}^\top (\mathbf{A}\mathbf{A}^\top)^{-1} (\mathbf{A}\mathbf{y} - \mathbf{b}).$$

Orthogonal Projection onto Hyperplanes

Consider the hyperplane

$$H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^\top \mathbf{x} = b\} \quad (\mathbf{0} \neq \mathbf{a} \in \mathbb{R}^n, b \in \mathbb{R}).$$

Then by the projection on an affine space result (above):

$$P_H(\mathbf{y}) = \mathbf{y} - \mathbf{a} (\mathbf{a}^\top \mathbf{a})^{-1} (\mathbf{a}^\top \mathbf{y} - b) = \mathbf{y} - \frac{\mathbf{a}^\top \mathbf{y} - b}{\|\mathbf{a}\|^2} \mathbf{a}.$$

Lemma (distance of a point from a hyperplane). Let $H = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^\top \mathbf{x} = b\}$, where $\mathbf{0} \neq \mathbf{a} \in \mathbb{R}^n$ and $b \in \mathbb{R}$. Then

$$d(\mathbf{y}, H) = \frac{|\mathbf{a}^\top \mathbf{y} - b|}{\|\mathbf{a}\|}$$

Proof.

$$d(\mathbf{y}, H) = \|\mathbf{y} - P_H(\mathbf{y})\| = \left\| \mathbf{y} - \left(\mathbf{y} - \frac{\mathbf{a}^\top \mathbf{y} - b}{\|\mathbf{a}\|^2} \mathbf{a} \right) \right\| = \frac{|\mathbf{a}^\top \mathbf{y} - b|}{\|\mathbf{a}\|}$$

□

Similarly, it follows the computation of the orthogonal projection onto half-spaces.

Lemma. Let $H^- = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^\top \mathbf{x} \leq b\}$ where $\mathbf{0} \neq \mathbf{a} \in \mathbb{R}^n$ and $b \in \mathbb{R}$. Then

$$P_{H^-}(\mathbf{x}) = \mathbf{x} - \frac{[\mathbf{a}^\top \mathbf{x} - b]_+}{\|\mathbf{a}\|^2} \mathbf{a}.$$

KKT conditions for nonlinear problems

Now, we extend the notion of KKT conditions to the general nonlinear case. We will start by giving an alternative notion of necessary optimality conditions in terms of **feasible descent directions**, which we introduce in what follows. Consider the problem

$$\min \quad f(\mathbf{x}) \quad \text{s.t.} \quad \mathbf{x} \in C \tag{G}$$

with $C \subset \mathbb{R}^n$ convex and f a continuously differentiable function over C . A vector $\mathbf{d} \neq \mathbf{0}$ is called a feasible descent direction at $\mathbf{x} \in C$ if



- i) $\nabla f(\mathbf{x})^\top \mathbf{d} < 0$, and
- ii) there exists $\varepsilon > 0$ such that $\mathbf{x} + t\mathbf{d} \in C$ for all $t \in [0, \varepsilon]$.

Using the notion of feasible descent directions, we reformulate the necessary optimality conditions as given in the following Lemma.

Lemma. *If $\mathbf{x}^*$ is a local optimal solution of (G); then, there are no feasible descent directions at $\mathbf{x}^*$.*

Now, we will extend this result to problems of the type.

$$\begin{cases} \min f(\mathbf{x}) \\ \text{subject to } g_i(\mathbf{x}) \leq 0, \quad i = 1, 2, \dots, m \end{cases} \quad (\text{NLP})$$

We introduce the notion of **active constraints**. We say the i -th constraint is active at $\tilde{\mathbf{x}}$ if $g_i(\tilde{\mathbf{x}}) = 0$, i.e., when the constraints are satisfied as equalities. Moreover, the set

$$I(\tilde{\mathbf{x}}) = \{i \in \{1, \dots, m\} : g_i(\tilde{\mathbf{x}}) = 0\}$$

is called the set of active constraints at $\tilde{\mathbf{x}}$.

Lemma. *Let $\mathbf{x}^*$ be a local minimum of (NLP), where f and $g_1, \dots, g_m$ are continuously differentiable functions over $\mathbb{R}^n$. Let $I(\mathbf{x}^*)$ be the set of active constraints at $\mathbf{x}^*$. Then, there does not exist a vector $\mathbf{d} \in \mathbb{R}^n$ such that*

- i) $\nabla f(\mathbf{x}^*)^\top \mathbf{d} < 0$, and
- ii) $\nabla g_i(\mathbf{x}^*)^\top \mathbf{d} < 0$, for all $i \in I(\mathbf{x}^*)$.

The previous theorem shows that a necessary optimality condition for local optimality is the **infeasibility** of certain system of strict inequalities. More treatable assumptions over the constraints can be imposed to guarantee the infeasibility of previous system, and they are commonly referred to in the literature as *constraint qualifications*. In the next section, we will show one constraint qualification which works in particular in the convex case.

KKT conditions for nonlinear convex problems

Let us start by recalling that for convex objective functions, the KKT conditions are always sufficient even in the nonlinear case. Therefore, we obtain the following result.



Theorem (Sufficiency of the KKT conditions for convex optimization problems). *Let $\mathbf{x}^*$ be a feasible solution of*

$$\begin{cases} \min f(\mathbf{x}) \\ \text{subject to } g_i(\mathbf{x}) \leq 0, \quad i = 1, 2, \dots, m \\ h_j(\mathbf{x}) = 0, \quad j = 1, 2, \dots, p, \end{cases} \quad (\text{NLP})$$

where $f, g_1, \dots, g_m$ are continuously differentiable convex functions over $\mathbb{R}^n$ and $h_1, \dots, h_p$ are affine functions. Suppose that there exist multipliers $\lambda_1, \dots, \lambda_m \geq 0$ and $\mu_1, \dots, \mu_p \in \mathbb{R}$ such that

$$\nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i \nabla g_i(\mathbf{x}^*) + \sum_{j=1}^p \mu_j \nabla h_j(\mathbf{x}^*) = 0, \quad (1)$$

$$\lambda_i g_i(\mathbf{x}^*) = 0, \quad i = 1, 2, \dots, m. \quad (2)$$

Then $\mathbf{x}^*$ is an optimal solution of (NLP).

A more refined result is stated for necessity of KKT conditions, that is, whether optimal solutions do satisfy the KKT system.

Theorem (necessity of the KKT conditions under the generalized Slater's condition). *Let $\mathbf{x}^*$ be an optimal solution of the problem*

$$\begin{cases} \min f(\mathbf{x}) \\ \text{subject to } g_i(\mathbf{x}) \leq 0, \quad i = 1, 2, \dots, m \\ h_j(\mathbf{x}) \leq 0, \quad j = 1, 2, \dots, p, \\ s_k(\mathbf{x}) = 0, \quad k = 1, 2, \dots, q, \end{cases} \quad (\text{NLP2})$$

where $f, g_1, \dots, g_m$ are continuously differentiable convex functions over $\mathbb{R}^n$ and $h_1, \dots, h_p$, and $s_1, \dots, s_k$ are affine functions. Suppose that there exists a point $\hat{\mathbf{x}}$ satisfying the generalized Slater's condition

$$g_i(\hat{\mathbf{x}}) < 0, \quad i = 1, 2, \dots, m \quad (3)$$

$$h_j(\hat{\mathbf{x}}) \leq 0, \quad j = 1, 2, \dots, p, \quad (4)$$

$$s_k(\hat{\mathbf{x}}) = 0, \quad k = 1, 2, \dots, q. \quad (5)$$

Then, there exist multipliers $\lambda_1, \dots, \lambda_m, \eta_1, \dots, \eta_p \geq 0$ and $\mu_1, \dots, \mu_q \in \mathbb{R}$ such that

$$\nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i \nabla g_i(\mathbf{x}^*) + \sum_{j=1}^p \eta_j \nabla h_j(\mathbf{x}^*) + \sum_{k=1}^q \mu_k \nabla s_k(\mathbf{x}^*) = 0, \quad (6)$$

$$\lambda_i g_i(\mathbf{x}^*) = 0, \quad i = 1, 2, \dots, m, \quad (7)$$

$$\eta_j h_j(\mathbf{x}^*) = 0, \quad j = 1, 2, \dots, p. \quad (8)$$

